TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Density of Water at Various Temperatures by a Specific Gravity Bottle

A bottle that always holds the same volume turns density into three weighings — and shows water's density falling, with its famous anomaly near 4 °C.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 7 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to measure the density of water at several temperatures. A **specific gravity (density) bottle** is a small flask with a perforated stopper: filled and stoppered, the excess liquid escapes through the capillary hole, so the bottle always contains the **same volume** V. Weigh it empty (W_0), filled with water at the reference temperature 4 °C (W_4), and filled at the temperature t under study (W_t). The water masses are then in the ratio of the densities:

$$\rho_t = \left((W_t - W_0) / (W_4 - W_0) \right) \times \rho_4, \quad \rho_4 = 1000 \text{ kg m}^{-3}$$

The subtlety that makes the experiment honest: the bottle itself expands slightly with temperature, changing V; a correction using the glass's cubical expansion coefficient γ ($V_t = V_4(1 + \gamma(t - 4))$) is applied in careful work, as shown in the model calculation.

2. Apparatus and procedure

- **Apparatus:** specific gravity bottle (25 or 50 mL), balance (to 0.001 g if available), thermometer, water baths at several temperatures (ice-water for low, thermostat/warm baths for high), distilled water, blotting paper.
- Clean and dry the bottle; weigh empty. Fill with freshly boiled distilled water (boiling expels dissolved air), equilibrate in the 4 °C bath (ice-water adjusted), insert the stopper, wipe the overflow dry, weigh.
- Repeat the filling-and-weighing with the water equilibrated at each chosen temperature (e.g. 10, 20, 30, 40, 50 °C): the water and the bottle must both reach the bath temperature before stoppering.
- Compute ρ_t at each temperature (with the glass-expansion correction) and plot ρ against t.

3. Worked calculation (model data)

RESULT AT 30 °C

W_0 = 15.240 g; W_4 = 40.240 g → water at 4 °C = 25.000 g (V_4 = 25.000 mL). At 30 °C: W_{30} = 40.129 g → water mass = 24.889 g.

Uncorrected: $\rho_{30} = (24.889/25.000) \times 1000 = 995.6$ kg m^{-3}.

Glass correction (γ_glass ≈ 2.5×10^{-5} K^{-1}): ΔT = 26 K → $V_{30} = V_4(1 + 2.5 \times 10^{-5} \times 26) = V_4 \times 1.00065$. True density = 995.6/1.00065 = **994.9 kg m^{-3}**.

The accepted value at 30 °C is 995.7 kg m^{-3}; the uncorrected school result lands closer because a second small effect (water retained/expansion of the neck contents) partly compensates — the experiment teaches that "density of water" numbers at the 0.1% level are arguments about corrections, and the raw trend is the real content: ρ falls steadily from its 4 °C maximum (1000) through ≈ 998.2 (20 °C) to ≈ 988.0 (50 °C).

4. Precautions

- No air bubbles: fill by pouring down the wall, and use boiled (degassed) water — a bubble is volume that is not water.
- The outside of the bottle must be perfectly dry at weighing, and the bottle handled by its neck (finger warmth expands the contents and squeezes liquid out of the overflow).
- Temperature equilibrium is the slow step: allow several minutes in the bath and confirm with the thermometer in the bath, not the room.
- Weigh promptly after wiping — evaporation from the capillary hole is continuous.
- The stopper must seat identically every time; mark its orientation.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is water at 4 °C the natural reference for this bottle?
2. With W_0 = 18.320 g, W_4 = 43.320 g, W_{50} = 42.995 g, find ρ_{50} uncorrected.
3. Why must the water be freshly boiled before filling?
4. State the direction of the glass-expansion correction and why it lowers the computed density at high temperature.
5. The plotted ρ–t curve is nearly flat between 0 and 8 °C and steepens later. What physical anomaly does the flat top show?

6. Why handle the filled bottle by the neck only?

Answer key

1. Because water's density is maximum at 4 °C (1000 kg m^{-3} by definition of the classical scale), so the bottle's reference volume is calibrated against the densest, most reproducible state of the very liquid being studied.
2. Water at 4 °C: 43.320 − 18.320 = 25.000 g. Water at 50 °C: 42.995 − 18.320 = 24.675 g. ρ_{50} = (24.675/25.000) × 1000 = 987.0 kg m^{-3} — close to the accepted 988.0, the small deficit being typical of a trace of evaporation or incomplete warming.
3. Boiling expels dissolved air. Unboiled water releases micro-bubbles inside the bottle as it warms or stands; each bubble displaces water through the overflow and the weighed "water" mass comes out low.
4. At higher temperature the bottle's volume is larger, so the same weighed mass occupies a larger volume than the 4 °C calibration assumes; dividing by the larger true volume lowers ρ. (The correction grows with ΔT — negligible at 10 °C, worth about 0.7 kg m^{-3} at 30 °C, more at 50 °C.)
5. Water's anomalous expansion: between 0 °C and 4 °C water contracts on warming (density rising to its maximum at 4 °C) before normal expansion takes over. The flat, rounded top of the curve is that anomaly made visible — the reason ice floats and lakes freeze from the top.
6. Because hand warmth on the body of the bottle warms the water, which expands and expels liquid through the capillary overflow — the bottle then no longer holds its calibrated full volume when weighed.

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